Predicting Message Influence Before Acting: A Trust Filter for Multi-Agent Discom Decisions

PRATIK PANDA
60003250251

RUDRA BHATT
60003250107

YASH LAKHANI
60003250097

VANSH KOTECHA
60003250036

Under the guidance of

Prof. Monali Sankhe

Project Guide

Primary user: power distribution companies (discoms)

Tested by replaying real hourly data from BDG2: 18 sites, 1,578 buildings

Department of Information Technology
DJSCE, Mumbai University
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Introduction & Problem Statement

Operational setting: A discom buys power a day ahead and pays deviation prices for any gap between plan and actual demand. It also decides which meter or feeder alert deserves a field visit today. Both decisions rest on messages that may be wrong or stale.

Distributed evidence: Forecast updates, outage notices, theft and anomaly flags, and helper agents (fixed-limit, weather-adjusted, peer, trend, later solar) each send a message with a size and a confidence, and they often disagree. India's smart-meter rollout is multiplying them.

Core problem: A message can help, hurt or not matter, depending on the system's state. Checking every claim is too slow and costly; trusting every claim is unsafe. Under our assumed 2 to 3 times shortfall price and half-price resale, buying short costs 2 to 4 times more per unit than buying surplus.

Research challenge: The discom cannot tell, before acting, whether a message will improve, worsen or barely affect the decision. Existing tools forecast demand or flag anomalies; we found none that predicts the value of acting on the message itself.

Research question: Can pre-decision prediction of message influence filter untrustworthy evidence and cut purchase-error cost and wasted inspection visits, without adding verification cost?

What a wrong message costs a discom in India

₹5.39 per kWh
average deviation price, 2022-23 (₹3.73 a year earlier)

₹40 per kWh
deviation charge in the worst blocks of 2022-23

₹12+ per kWh
Delhi evening peak, 5 to 9 PM, 2021

₹2.16 per kWh lost
on surplus sold off-peak, Rajasthan

15.04% AT&C loss, FY25
down from 21.91% in FY21; 5.83 crore smart meters installed by Feb 2026 against a 25 crore target

Sources: CERC Annual Report on Short-Term Power Market 2022-23; CEEW study on Delhi procurement; IEEFA, The Curious Case of India's Discoms (2020); Ministry of Power reply in Lok Sabha (2025); Parliamentary Standing Committee report (2026). The 2 to 4 times figure is derived from assumed prices, not measured.

Proposed Methodology

Objective: Predict, before a decision is made, whether a candidate message is Beneficial, Harmful or Negligible for a discom's purchase or inspection decision, using replay on real hourly meter and weather data.

Mapping to a discom: Each BDG2 site acts as a feeder area and each building as a large consumer on it. BDG2 is non-residential US and European data, so we state this as a limitation.

Stage 1: Prepare: Load meter, weather and metadata; keep buildings with over 80% of readings; interpolate short gaps; scale each building by its own mean.

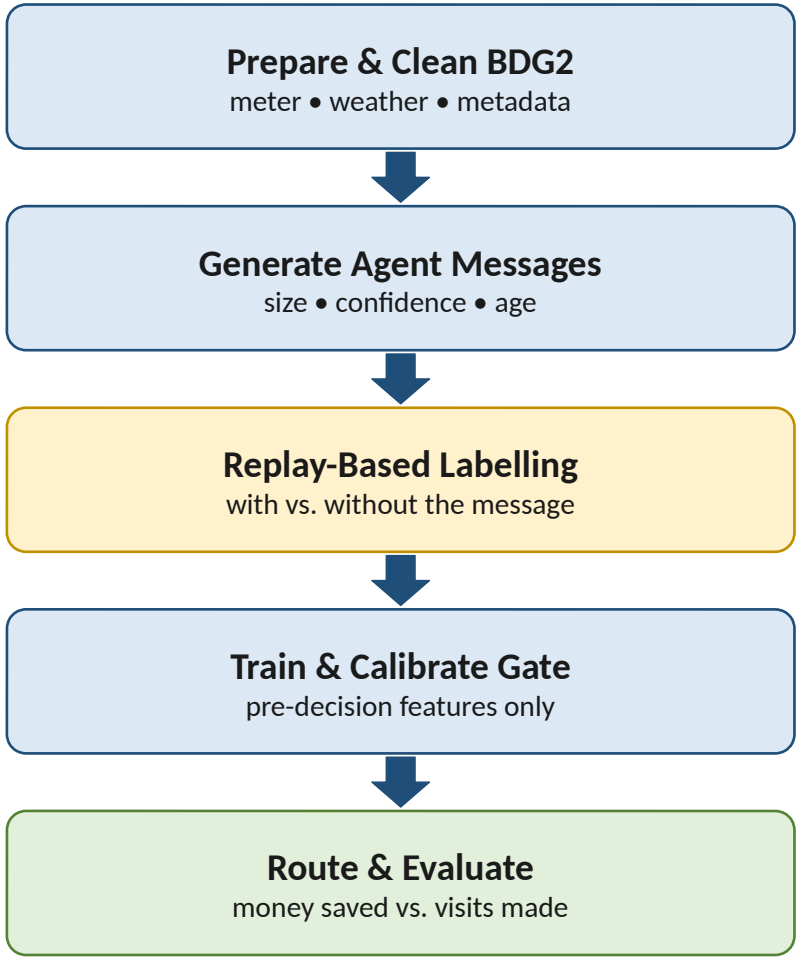
Stage 2: Generate: A forecaster and rule-based helper agents emit messages with a size, a confidence and a timestamp. Natural false alarms are kept and known faults are injected.

Stage 3: Label by replay: Replay each message with and without it against realised load. Inspection value = extra kWh that persisted over 24 to 72 hours × tariff − visit cost. Planning value = purchase cost with versus without the message, using shortfall and resale prices.

Stage 4: Learn: A LightGBM gate estimates three influence probabilities from pre-decision features only; isotonic calibration makes them usable.

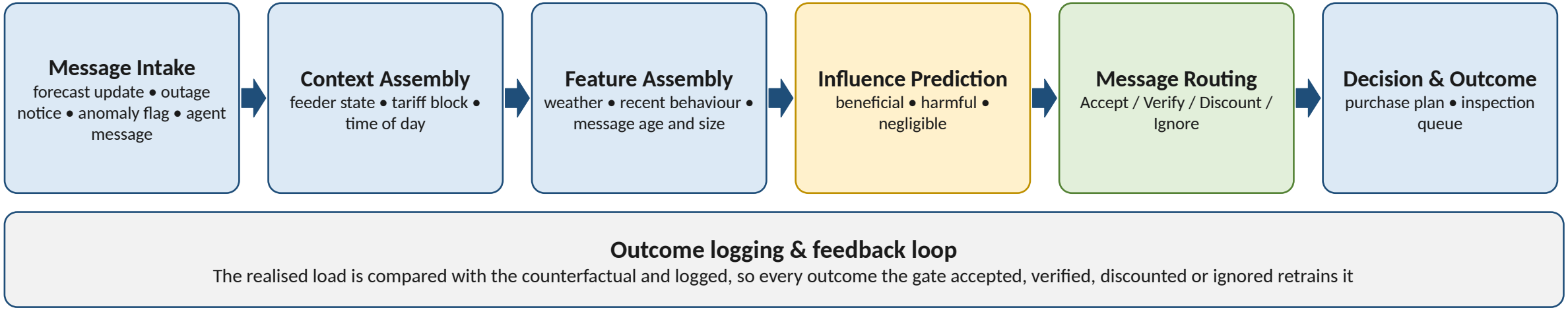
Stage 5: Route & evaluate: Predicted influence selects Accept, Verify, Discount or Ignore under an asymmetric cost, scored on money saved against visits made.

Methodology Pipeline



Process Flow

Message-to-decision process with influence-aware routing



Practice today, from discoms' own words

Punjab (PSPCL): told CERC that inaccurate real-time data raised its deviation-charge burden.

Adani Electricity Mumbai: says discom deviations come largely from demand swings it cannot control.

MP discom (MPPoKVVCL): analyses each meter event in isolation and asks for AI to judge combined severity.

India Smart Grid Forum: says field teams face thousands of daily alerts and proposes risk-based inspection.

What the gate changes

Before: every message is acted on, or a fixed threshold decides.

After: each message gets a predicted value and one of four actions.

Same decision makers: purchase planners and inspection teams, with no extra verification step by default.

Sources: CERC draft-regulation comments (PSPCL 2019, AEML 2024); IIT Bombay SINE problem statements; India Smart Grid Forum presentation (ADB, 2026). Discom or conference statements, not audited measurements.

Selected Algorithms / Techniques / Methodologies

Component	Technique / Algorithm	Purpose & Rationale
Operational simulation	Replay of real BDG2 hourly data with injected faults	Keeps every episode reproducible and grounded in realised load, not a synthetic demand model.
Forecaster	Same-hour-last-week baseline versus LightGBM regression	Supplies expected load for the agents; the naive baseline is the bar every model must beat.
Message generation	Rule-based helper agents: fixed-limit (25% above normal), weather-adjusted, peer, trend	Produces supporting, conflicting and redundant messages across independent agents without needing AI first.
Ground-truth labelling	Counterfactual replay, with versus without the message	Labels influence as Beneficial, Harmful or Negligible objectively, with no manual annotation.
Influence prediction	LightGBM gradient-boosted trees (baseline: logistic regression)	Models mixed pre-decision features and stays interpretable through feature importance.
Probability calibration	Isotonic regression (CalibratedClassifierCV)	Makes the three class probabilities usable for routing thresholds (low ECE).
Routing policy	Expected-utility threshold rule with asymmetric cost	Balances money saved against visits and verification cost, penalising under-buying more than over-buying.
Baselines to beat	Always-act, never-act, fixed threshold, the agent's own confidence, disagreement rule	Shows whether predicting influence adds value beyond simple rules.

Dataset / Preprocessing / Implementation

Dataset: Building Data Genome 2 (BDG2): 18 site folders, 1,578 buildings in the full meter file, hourly from 1 Jan 2016 to 31 Dec 2017 (17,544 rows), three files per site: meter, weather, metadata.

Structure: meter.csv: one row per hour, one column per building, in kWh, the realised load every decision is scored against. weather.csv: hourly air temperature, dew point, pressure, wind per site. metadata.csv: one row per building (usage type, sqm, coordinates, year built).

Data quality: About 9% of meter values are missing. cloudCoverage is 52% empty and precipDepth6HR 95% empty, so both are dropped. Site Shrew is 45.6% missing and Panther 19.8%, so we avoid or check them.

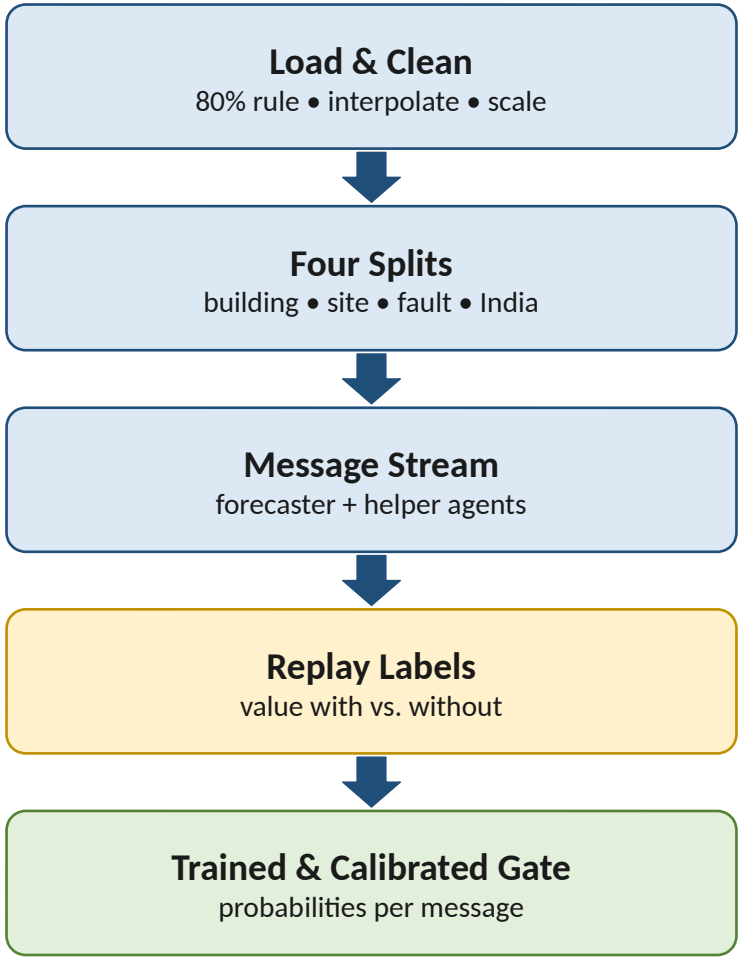
Feature groups: Building context, time, weather, recent behaviour, and message attributes: size relative to normal, age, and which agents agree.

Preprocessing: Keep buildings with over 80% of readings; interpolate gaps of a few hours; leave long gaps blank; scale each building by its own mean because sizes run from 0.02 to 3,400 kWh.

Leakage control: Only pre-decision features are kept, checked by timestamp. Splits are by building or site, so blanks and future hours never leak across train and test.

Implementation: Python: pandas, NumPy, scikit-learn, LightGBM, Jupyter. Developed on one site (Robin, 51 buildings), then scaled to the full file.

Data & Implementation Pipeline



Inside BDG2: the files we use most (real Robin rows)

Meter file: one row per hour, one column per building (kWh). First rows of site Robin.

hour	Robin_pub...	Robin_lodg...	Robin_edu...	Robin_offic...	Robin_edu...
1	36.438	14.3	52.25	13.378	229
2	70.75	27.3	106.25	26.755	449.5
3	74.312	25.7	104.25	26.755	451.5

Weather file: one row per hour per site. First rows of site Robin.

hour	airTemp	cloud	dewTemp	seaLvlPressure	windDir	windSpeed
1	3.8	blank	2.4	1020.9	240	3.1
2	3.7	0	2.4	1021.6	230	2.6
3	2.6	0	1.9	1021.9	0	0

Metadata file: one row per building.

building_id	primaryspaceusage	sqm	yearbuilt
Robin_pub...	Public services	10984	blank
Robin_lodg...	Lodging/residential	5842.9	1960
Robin_edu...	Education	6337	1976

Source: our downloaded BDG2 files for site Robin (London, 51.52°N). Column names are shortened in the file view; blank means no value in the file.

How the gate uses each file

Meter (truth): kWh is what each consumer really used. It is the ground truth for replay labels: every accept, verify, discount or ignore is scored against it.

Weather (helper input): Temperature and wind feed the weather-adjusted agent and the weather feature group. Cloud and 6-hour precipitation are too sparse.

Metadata (context): Usage type, size and site: the grouping key for splits and a context feature, so an office and a dormitory are treated differently.

In discom terms: A site is a feeder area; a building is a large consumer on it.

Research Questions & Hypotheses

Main question: Can the value of acting on a message be predicted before acting, and does gating on it cut wasted inspections and purchase-error cost versus always-act, fixed-threshold and confidence baselines?

Research question	Hypothesis (a target to test, not a finding)	Measured by
RQ1: What share of messages are beneficial, harmful or negligible?	H1: a large share of raw messages (we guess 40 to 60%) turn out negligible or harmful	Share of each label from replay
RQ2: How well do pre-decision features predict that, and are the predictions calibrated?	H2: pre-decision features predict value better than the agent's own confidence	Macro-F1, AUROC, calibration error (ECE)
RQ3: Does gating give a better money-versus-effort trade-off?	H3: gating cuts wasted visits by about 20% or more at the same recovered value, versus always-act. H5: an asymmetric-cost gate beats a symmetric one once the shortfall multiple is 2× or more	Cost saved against visits made, across the price grid
RQ4: Does it hold on unseen buildings, sites, fault types and Indian data?	H4: the gain shrinks on unseen sites but stays positive	Four splits, including the I-BLEND India test

The percentages and the 2 times multiple are our own guesses, set as targets; the experiments will confirm or reject them. RQ1 to RQ4 map to H1 to H5 as shown.

Implementation Plan

#	What we do	Output
1	Load and clean: keep buildings over 80% complete, interpolate short gaps, scale by building mean	Clean tables
2	Fix four splits before modelling	Train and test lists
3	Forecaster: same-hour-last-week versus LightGBM on time, weather, size	Forecast and its error
4	Message agents: fixed-limit, weather-adjusted, peer, trend	Message stream
5	Fault injection: stuck, spike, drift, offset on held-out periods, plus natural false alarms	Cases with known truth
6	Replay labels: shortfall multiple 1.5× to 3×, resale 0.3 to 0.7, visit cost	Beneficial / harmful / negligible
7	Features: context, time, weather, behaviour, message size, age, agreement; timestamp check	Training table
8	Train LightGBM, calibrate isotonic, five seeds	Three probabilities
9	Baselines and routing: cost-minimising thresholds on a validation set	Accept / Verify / Discount / Ignore
10	Evaluate: macro-F1, ECE, money saved against visits, ablations	Result tables, plots

The four splits

Building: 80% train, 20% test, mixed by usage and site (RQ2, RQ3)

Site hold-out: train on all but one site (RQ4)

Fault hold-out: withhold one fault type (RQ4)

India transfer: train on BDG2, test on I-BLEND (RQ4)

Suggested 8-week timeline

Weeks 1 to 2: steps 1 to 3

Weeks 3 to 4: steps 4 to 6

Weeks 5 to 6: steps 7 to 9

Weeks 7 to 8: step 10 and write-up

Tools: Anaconda, Jupyter, pandas, scikit-learn, LightGBM. Never split by time inside one building: the same building would leak into both sets. Timeline is a suggestion; adjust to the real deadline.

Evaluation & Expected Outcomes

Influence prediction: Macro-F1 and AUROC over the three classes, with calibration error (ECE) and reliability curves.

Operational impact: Purchase-error cost under the shortfall multiple grid (1.5×, 2.5×, 3×) and resale fraction (0.3 to 0.7), wasted field visits, faults recovered sooner and alarms per day, against a no-gate baseline.

Cost of verification: Verification calls, messages acted on and latency, as a utility-per-cost ratio.

Ablation: Always-act, gate without routing, full pipeline; drop one feature group or one agent at a time.

Expected outcome: Fewer harmful messages acted upon and a better money-versus-effort trade-off than unselective acting; the gain shrinks on unseen sites but stays positive.

Scale (illustrative): A mid-size discom with about ₹120 crore a year of deviation cost would save about ₹12 crore from a 10% cut. Every input is assumed.

What would count as success or failure (proposed)

- Success: the gate beats always-act and the fixed threshold on money saved against visits made in at least 3 of the 4 splits.
- Success: calibration error stays low enough to set routing thresholds.
- Failure: no gain over the agent's own confidence, or the gain disappears on an unseen site.
- A negative result is still reported; it shows where the gate is not useful.

Limits we state openly

- BDG2 is non-residential US and European data, not Indian households.
- Messages are partly simulated; "worth acting on" is a proxy label.
- Prices and the 2 to 3× multiple are assumed, so we report a grid.
- No real discom purchase data yet; this is a replay study.

Conclusion

Contribution: A framework that predicts message influence before a decision and routes messages accordingly, so beneficial evidence is used and misleading evidence is filtered out.

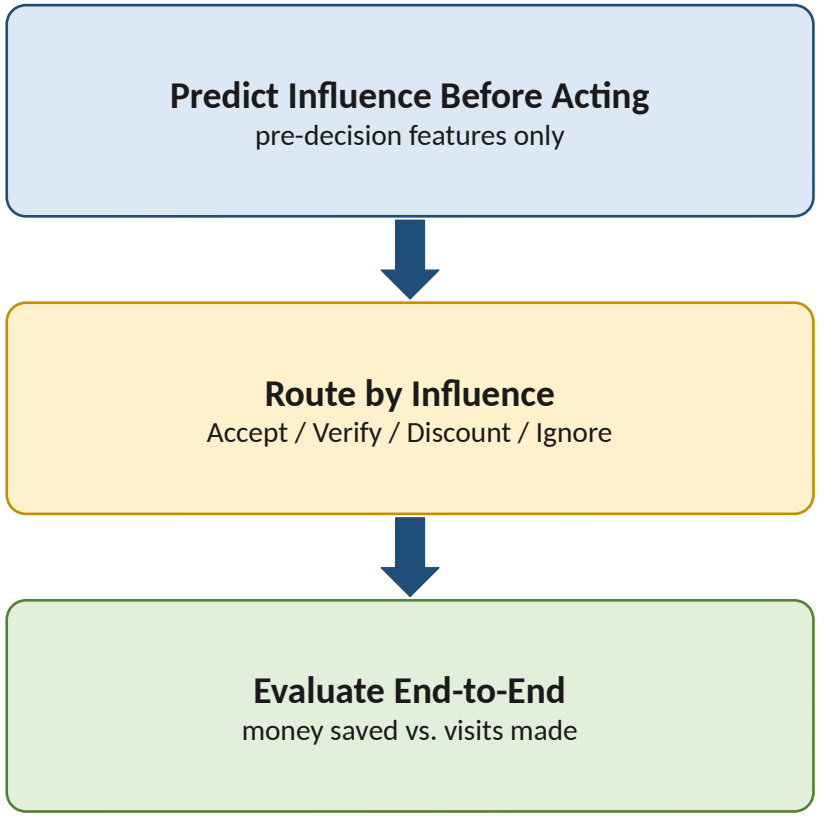
Method: Replay-based labelling on real hourly data, pre-decision prediction with a calibrated LightGBM gate, and influence-aware routing under an asymmetric cost.

Evaluation: Connects message-level influence to purchase cost and field effort, rather than judging messages in isolation or counting them.

Impact: A discom can accept, verify, discount or ignore each incoming message: fewer penalty buys, fewer wasted visits. Facility managers and solar or storage operators gain the same way.

Future work: India transfer test on I-BLEND, solar-drop and battery agents, and a pilot that replays one real feeder or a month of purchase data to replace the assumed prices.

What This Work Delivers



Thank You!!!

Predicting message influence before acting: a trust filter for multi-agent discom decisions